Research notes & sources

Background behind the tracker: paper reviews, IRB mapping, the chatbot QA test, and the Mexico-data-ethics discussion.

Shared papers: applicability

Foundation Models for Biomedical Image Segmentation: A Survey reviews roughly 100 papers (Apr–Sep 2023) adapting Meta's Segment Anything Model (SAM) to medical imaging across 33 public datasets, mostly zero/few-shot. SAM "approaches or achieves state-of-the-art performance" on many structures but is documented as weak specifically on the carotid artery, adrenal glands, optic nerve, and mandible bone. The optic-nerve weakness is a direct caution for a glaucoma-imaging pivot: cup-to-disc ratio assessment depends on accurate optic-nerve-head segmentation, exactly where general-purpose foundation models currently struggle most.

nnU-Net (Isensee et al., Nature Methods 2021, DKFZ/Heidelberg) is a self-configuring segmentation framework: given a new dataset, it automatically picks preprocessing, network architecture, and training schedule without manual tuning, and has won or matched top results across dozens of public biomedical segmentation challenges spanning CT, MRI, and other modalities.

Applicability: both papers point to the same practical strategy for a team with no funding and no data of its own: don't train a segmentation model from scratch. Fine-tune a foundation model, or configure nnU-Net, on whatever small public dataset you can get for free. This directly answers the "how do we get started at low/no cost" question from the last meeting, though optic-nerve/glaucoma structures specifically may need extra validation given SAM's documented weakness there.

IRB checklist: what it is and what it means for us

IRB (Institutional Review Board): a standard IRB submission packet (eligibility questionnaire, informed consent, SOPs, CITI training certificates, etc.). Two practical implications:

1. **Using existing public datasets** (AI-READI, EyePACS, CP-AnemiC, TCIA, etc.) does not require a new IRB protocol from your team.
   a. original data collectors already went through their own IRB and consent process
   b. fastest path and needs none of the IRB checklist
2. **Running your own study** (recruiting participants, collecting new eye photos or A1c readings, testing the device on people) does need the full packet Dr. Penkova posted
   a. eligibility screening, informed consent, CITI training (Biomedical, GCP, HIPAA) for everyone touching data
   b. SOPs for device use and sample handling
   c. Budget real time for this: IRB review, even expedited, typically takes weeks, and CITI certificates must be completed before anyone on the team can touch participant data

Mexico data-collection idea

Our meeting notes mention Mexican clinics having "more relaxed rules" as a reason to collect data there. Flagging this clearly: pursuing a jurisdiction because its human-subjects oversight is weaker is a recognized research-ethics problem sometimes called "ethics dumping," and it carries real practical risk beyond the ethics itself:

- IRB jurisdiction follows the researchers' and institution's involvement in a study, not just where data collection physically happens
  - USC-affiliated research on human subjects abroad still typically needs USC IRB review (often via a reliance agreement with a local ethics committee), regardless of local rules.
- NIH and most foundations require equivalent human-subjects protections for international sites, not reduced ones.
- Journals increasingly screen for exactly this pattern during peer review
  - can sink a paper or a grant even after the data is already collected

International collaboration itself is common and legitimate in NIH-funded research (PAR-25-357 explicitly allows foreign organizations).

- **Solution:** pursuing the mexico data capture angle is viable given:
  - its through a local ethics committee
  - USC IRB reliance
  - proper informed consent, the same as any other site, rather than treating relaxed local rules as the selling point

DoD / DARPA / ARPA-H angle

- **DARPA Biological Technologies Office**: rolling abstract submissions through Sep 30, 2026, but scoped to warfighter protection, tactical care, and chemical/biological threats. Anemia is the best fit: non-invasive, rapid anemia/blood-loss screening is directly relevant to combat casualty triage. Diabetes/glaucoma consumer screening is a weaker fit for this office's mission.
- ARPA-H (Advanced Research Projects Agency for Health), the "DARPA for health" agency. Its new Delphi program (launched April 2026) funds wearable/ingestible biosensors for continuous chronic-disease monitoring
  - conceptually close to the A1c-monitoring idea, though Delphi's specific hardware focus (implanted/wearable chiplets) doesn't match our camera-based approach
  - worth tracking ARPA-H's broader solicitation pipeline for a better-matched future call.
- CDMRP Vision Research Program looked promising on paper (up to ~$2.6M)
  - scoped to military-exposure-related vision injury specifically
  - FY26 is already invite/LOI-only
- **Solution:** pitch anemia to DARPA/DHA as triage screening, not as a consumer wellness app; pitch diabetes to NIH/ADA as health equity/access, not as a monitoring gadget.

Glaucoma chatbot — live QA test (2026-09-16)

Tested glaucoma.biotransportai.com directly with the two example questions highlighted in prior meetings:

Question asked	Result
"How do I put in eyedrops?"	Failed — "Something went wrong while generating a response. Please try again."

Question asked	Result
"Can I go swimming?"	Failed — same error

2 of 2 test questions failed with a generic backend error, not a content/knowledge gap.

Before advertising, check the LLM API key/quota behind the Streamlit app and the app's server logs (Streamlit Community Cloud's log panel, if that's the host)

This looks like a backend/config issue rather than a prompt-quality issue.

Potential collaborators for the paper

- NEI + NIBIB's "Eyes on Health" partnered research study (oculomics)
- A named multi-site consortium already working in oculomics-AI: Penn Medicine, Penn Engineering, University of Michigan Kellogg Eye Center, St. John Eye Hospital (Jerusalem), and Gyeongsang National University (Korea)
- The AI-READI / Bridge2AI consortium: we'll likely be citing/using their data anyway, so direct engagement is a natural next step
- The authors of the two papers Dr. Penkova shared (DKFZ/Heidelberg for nnU-Net; the SAM-survey authors): a warm outreach angle since the papers are already in our working set

Hosting notes

AWS, GCP, and Azure all run research-credit programs (NCI's Imaging Data Commons already grants free GCP credits -> see the Funding tracker).

For anything touching real participant data, use a private VPC with encryption at rest and a signed Business Associate Agreement (all three clouds offer HIPAA-eligible services) rather than building custom infrastructure.

Sources

- PAR-25-357 (NIH NEI R21)
- BrightFocus National Glaucoma Research
- Shaffer Grant
- Diabetes Research Connection

- ADA Type 1 Diabetes RFA
- ASH Foundation
- NCI Imaging Data Commons
- DARPA Biological Technologies Office
- ARPA-H Delphi program coverage
- CDMRP Vision Research Program
- AI-READI / FAIRhub
- CP-AnemiC / anemia conjunctiva detection research
- TCIA data usage policy
- Foundation Models for Biomedical Image Segmentation: A Survey
- nnU-Net